

S&P GLOBAL · CRISIL HACKATHON 2026

RiskPulse

From financial news to portfolio stress decisions.

ABHI PANDEY · VIT BHOPAL UNIVERSITY
B.TECH CSE (AI & ML)

THE GAP

Financial risk moves faster than reporting cycles

The signal appears in text first. The decision arrives later.

- Signals surface early
Critical clues appear in news, regulatory updates, and market commentary.
- Teams work apart
Analysts read text while portfolio teams work in separate risk models.
- Handoffs slow action
Manual transfer delays the choice to hedge, investigate, or stress-test.
- RiskPulse reconnects them
The qualitative signal becomes a quantitative portfolio view.

RiskPulse turns unstructured text into an auditable risk signal

1. Collect multi-source news and sentiment feeds.
2. Clean text and normalize entities.
3. Score sentiment, classify the event, and calculate impact.
4. Trigger stress testing when impact reaches 7 or higher.
5. Surface the result in one risk dashboard.

DETERMINISTIC RULES KEEP THE REASONING
TRANSPARENT AND REPEATABLE.

Six features built for a fast risk review



Multi-source intake

News, social sentiment, and optional public RSS.



Interpretable sentiment

Scores range from -1.00 to +1.00.



Eight event classes

Financial events are grouped into eight categories.



Impact tiers

Impact runs 1–10 across Low, Medium, High, and Severe.



Portfolio stress test

Module B turns a severe signal into portfolio impact.



Review modules

Signal feed, NLP sandbox, and dashboard results.

The decision path is one connected workflow

News Input → Text Analysis → Event Classification → Risk Score → Stress Testing → Results Dashboard

High-impact gate

Impact Score ≥ 7 .

Structured handoff

Each stage leaves a structured signal for the next.

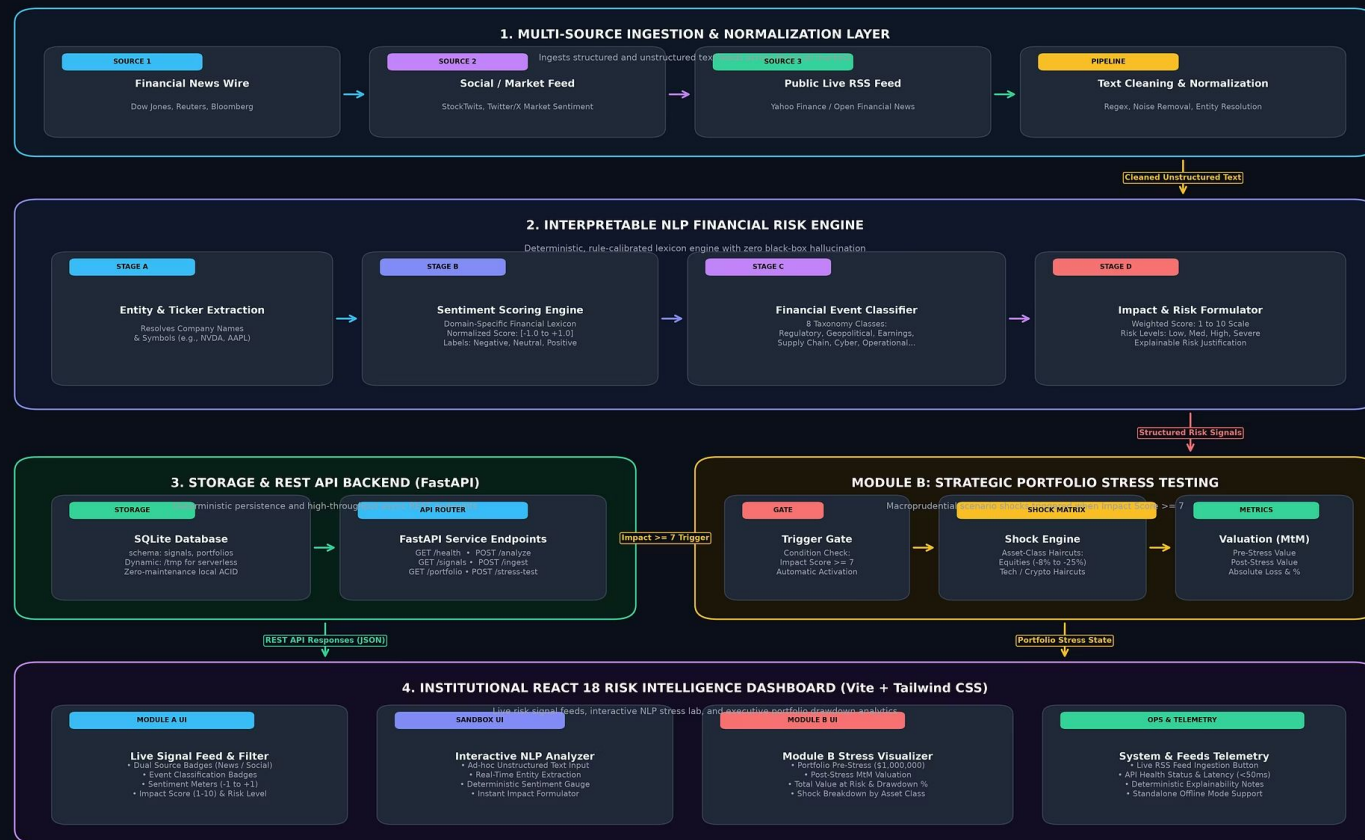
PROTOTYPE ARCHITECTURE

A lightweight stack keeps the prototype explainable

Feeds become structured signals through a small, inspectable path.

RiskPulse: AI/NLP Financial Risk Intelligence & Strategic Stress Testing

End-to-End System Architecture | S&P Global & CRISIL Campus Hackathon 2026 | Candidate: Abhi Pandey (VIT Bhopal)

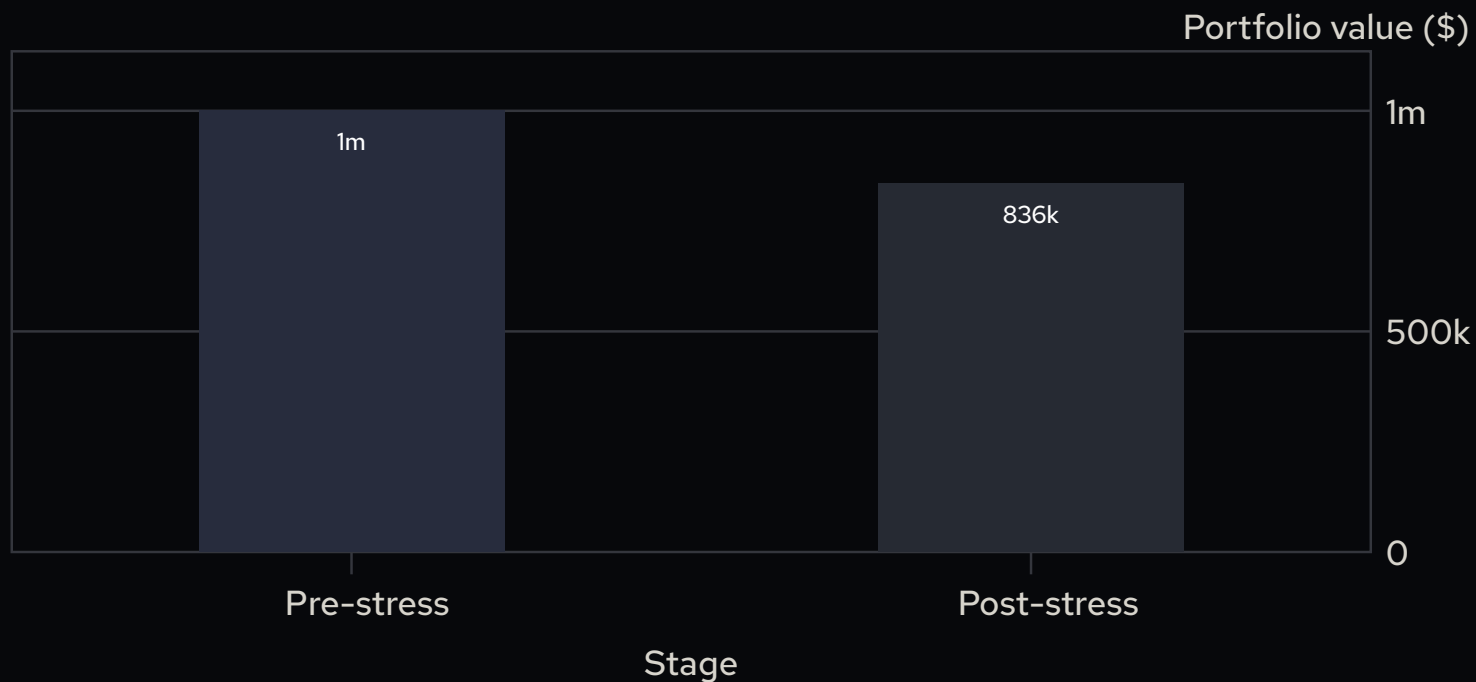


Built for S&P Global & CRISIL Campus Hackathon 2026 | Verified 30/30 Test Suite Passed | Deterministic & Fully Explainable

SYNTHETIC STRESS SCENARIO

A severe signal makes the portfolio impact visible

Regulatory event · Impact 8/10 · Module B activated



PORTFOLIO VIEW

\$164,000

Simulated loss

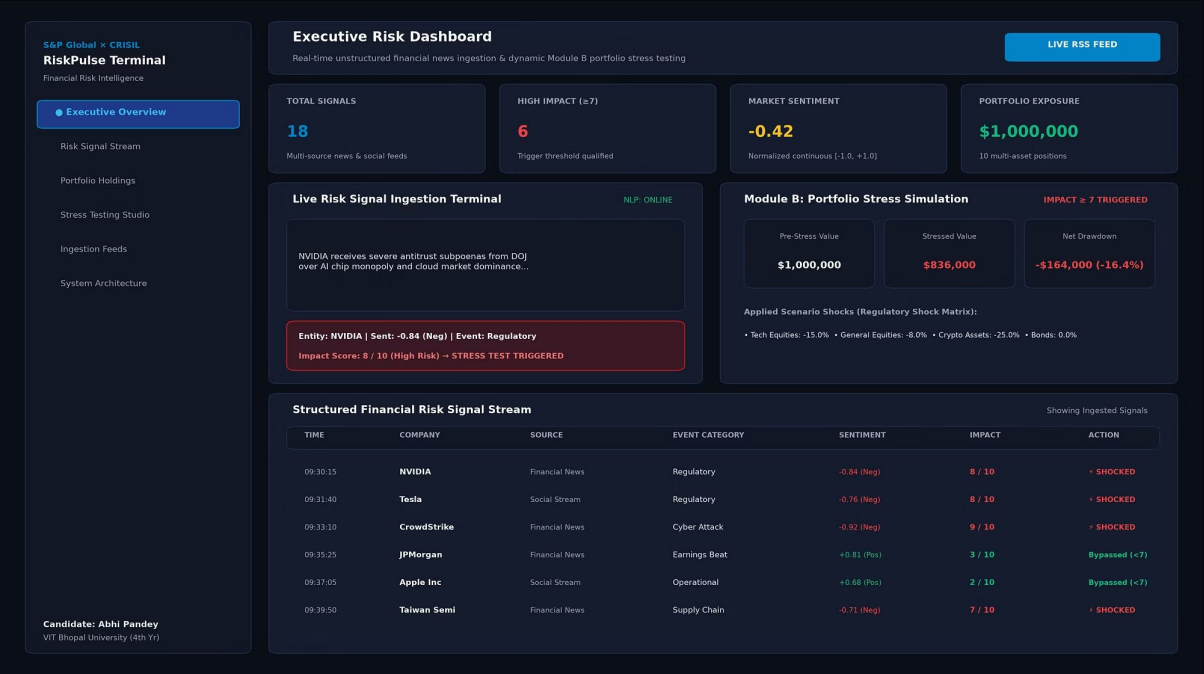
Baseline: \$1,000,000

Post-stress: \$836,000

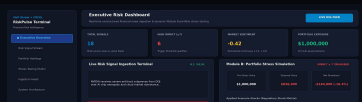
Drawdown: -16.4%

ACADEMIC BENCHMARK, NOT LIVE
CLIENT POSITIONS OR INVESTMENT
ADVICE.

Repository evidence shows the dashboard in action



EXECUTIVE DASHBOARD • SIGNALS, GATE, SENTIMENT, EXPOSURE



SENTIMENT GAUGE



STRESS HEATMAP

Executive view

Signal count, high-impact gate, market sentiment, and exposure.

Signal stream

Company, source, event, sentiment, impact, and action.

Module B

Pre-stress, stressed value, drawdown, and applied shocks.

VALIDATION AND LIMITS

The prototype is tested, measured, and honest about its limits

VERIFIED NOW

30/30

Automated tests passed

In the repository test run. Coverage: API routes, ingestion and RSS fallback, NLP scoring, portfolio math, and Module B.

MEASURED IN THE README

<15 ms

Average NLP inference latency

Synthetic maximum drawdown: -16.4%.

LIMITS

What remains bounded

Rule-based parsing, synthetic portfolio scope, fixed haircuts, and third-party RSS availability.

NEXT

Where it can grow

Hybrid FinBERT analysis, broker APIs, historical crisis backtesting, and VaR / Expected Shortfall.

THE TAKEAWAY

Detect. Analyse. Stress-Test. Decide.

RiskPulse links breaking financial text to an interpretable risk signal, then carries high-impact events into a portfolio stress view.

TRANSPARENT, STUDENT-READY PROTOTYPE · S&P GLOBAL & CRISIL HACKATHON 2026

GitHub: <https://github.com/Abhi4621/VIT-Bhopal-Abhi-Pandey-S-P-Crisil-hackathon>

Live demo: <https://vit-bhopal-abhi-pandey-s-p-crisil-hack.vercel.app/>

The supplied live-demo URL returned a Vercel 404 during verification; confirm the deployment before presenting it as live.

